

CS-499: Computer Science Capstone

Keagan Ryder

December 2025

The artifact I selected for my database enhancement is my Android Inventory Management Application originally created in CS-360: Mobile Architectures and Programming. This project was designed as a native Android application written in Java that relied on SQLite for storing user accounts, login credentials, and inventory item information. While the original version met the basic functional requirements of the course and successfully implemented simple CRUD operations and user authentication, it did not reflect the level of database structure, security, and engineering best practices expected from professional software development. For this milestone, I revisited the entire database component of the project and enhanced its schema, logic, and security so that it demonstrates stronger competency in database design and implementation.

I chose to include this artifact in my ePortfolio because it provides a meaningful demonstration of my ability to work with relational databases in a practical, real-world context. Databases are central to nearly every modern software system, and the ability to design them securely and efficiently is an essential skill for a computer science professional. The enhancements I made to this project showcase my abilities in this area far more effectively than the original version. In revisiting the artifact, I strengthened the database schema, improved referential integrity, eliminated redundant structures, and implemented secure handling of user credentials through salted password hashing. I also rewrote all SQL interactions to use prepared

statements, which protect the system from SQL injection vulnerabilities and reflect more advanced defensive programming techniques. Additionally, I restructured the database access layer to be more modular, organized, and maintainable. Error handling is now explicit and predictable, which makes the system more reliable and easier to extend. Performance improvements such as indexing and optimized queries were added to ensure that the application can scale beyond small test datasets. These enhancements transform the artifact from a basic academic exercise into a more robust and secure mobile application suitable for professional review.

In terms of course outcomes, this enhancement allowed me to make substantial progress toward the program objectives established in Module One. The work aligns strongly with the outcome related to software engineering and database skills, as the updated artifact demonstrates my ability to apply industry-standard techniques and tools in the design and implementation of a computing solution. The enhancement also directly supports the security mindset outcome. By redesigning the way user credentials are stored and by improving input validation and SQL protections, I showed an ability to anticipate adversarial threats and mitigate potential exploits in the architecture of the application. Although the algorithmic problem-solving outcome is not the primary focus of this enhancement, I applied algorithmic reasoning when redesigning query structures and adding indexes to improve performance. I still consider my CS-300 artifact to be the primary demonstration of algorithmic principles, so no changes to my overall outcome coverage plan are necessary.

Reflecting on the process of enhancing this artifact, I realized how much my skills have grown since I first wrote the original code. One of the most important lessons I learned is that

good database design is foundational to the stability and longevity of a software system. The original schema technically worked, but when examined with more experience, it became clear that it lacked structure, consistency, and scalability. Redesigning the schema taught me the value of normalization and proper relational modeling. I also learned that security must be intentionally designed into the system from the beginning. Implementing password hashing, salting, and prepared statements reinforced the idea that even small applications require strong security practices. Another lesson came from improving the reliability of the system through proper exception handling. In real applications, databases fail in unpredictable ways, and having meaningful error handling is just as important as the functionality itself. Finally, performance considerations became more important as I worked through the enhancement. Adding indexes and optimizing query paths demonstrated how small changes can greatly improve efficiency, even in a local mobile database.

The primary challenges I faced during this enhancement included refactoring older code that had not been designed for future expansion. Cleaning up the database helper and reworking structural logic required careful planning so that the user interface and overall application flow were not disrupted. Another challenge was working within Android's constraints on SQLite migrations, which required me to think critically about how to upgrade the schema without breaking existing functionality. It was also challenging to balance the desire to significantly improve the artifact with the need to keep the enhancement reasonable in scope. Ultimately, these challenges strengthened the artifact and helped reinforce my understanding of database engineering concepts.

Overall, this enhancement meaningfully improved the quality of the artifact and strengthened the credibility of my ePortfolio. The application is now more secure, more maintainable, and more reflective of professional industry expectations. Completing this enhancement allowed me to demonstrate progress toward multiple program outcomes and provided an opportunity to apply the knowledge and skills I have accumulated throughout the Computer Science program. I believe this improved artifact presents a clear example of my ability to design, implement, and secure a database system, and it serves as a strong representation of my readiness to enter the professional field.